Math708 – Foundations of Computational Mathematics I

Fall 2011

Homework 1

1. Write down the Lagrange interpolation polynomial $p_1(x) \in \mathbb{P}_1$ for the function $f : x \to x^3$ using the points $x_0 = 0$, $x_1 = a$. Verify Theorem 1.7 in the lecture notes by direct calculation, showing that ξ is unique and has the value $\xi = \frac{1}{3}(x + a)$.

2. Show that for pairwise distinct real numbers $x_0, x_1, \cdots, x_n$, the corresponding Lagrange polynomials satisfy:
$$\sum_{i=0}^{n} L_i(x) \equiv 1;$$
$$\sum_{i=0}^{n} L_i(0)x_i^j = \begin{cases} 1, & \text{for} \quad j = 0, \\ 0, & \text{for} \quad 1 \le j \le n, \\ (-1)^n x_0 x_1 \cdots x_n, & \text{for} \quad j = n+1. \end{cases}$$

3. Given $n + 1$ support points $\{(x_i, y_i)\}_{i=0}^{n}$ and let $p \in \mathbb{P}_n$ be the corresponding interpolating polynomial. Prove
$$p(x) = \sum_{i=0}^{n} \frac{\omega_{n+1}(x)}{(x - x_i){\omega'}_{n+1}(x_i)} y_i$$
where $\omega_{n+1}(x) = (x - x_0)(x - x_1)\cdots(x - x_n)$.

4. Show that for given pairwise distinct support nodes $x_0, x_1, \cdots, x_n \in \mathbb{R}$ and support ordinates $f_0, f_1, \cdots, f_n \in \mathbb{R}$, the corresponding divieded difference satisfy
$$f[x_0, \cdots, x_n] = \sum_{i=0}^{n} \left[\frac{f_i}{\prod_{k=0,k\ne i}^{n} (x_i - x_k)} \right].$$

5. Provide an estimate of
$$\max_{x\in[-1,1]} |\omega_{n+1}(x)|$$
for $\omega_{n+1}(x) = (x-x_0)(x-x_1)\cdots(x-x_n)$, in the case $n = 1$ and $n = 2$, for a distribution of equally spaced nodes $-1 = x_0 < x_1 < \cdots < x_n = 1$.

6. Construct the minimax polynomial $p_2 \in \mathbb{P}_2$ on the interval $[-1, 1]$ for the function g defined by $g(x) = \sin x$.

7. Suppose that $(x_i, f^{(k)}(x_i))$ are given data with $i = 0, \cdots, n$ and $k = 0, \cdots, m_i$. Let $N = \sum_{i=0}^{n}(m_i + 1)$, prove that, if the nodes $\{x_i\}_{i=0}^{n}$ are distinct, there exists a unique polynomial $H \in \mathbb{P}_{N-1}$, such that
$$H^{(k)}(x_i) = y_i^{(k)}, \quad i = 0, \cdots, n, \quad k = 0, \cdots, m_i,$$
where $y_i^{(k)} = f^{(k)}(x_i)$.

8. (**Programming Exercise**) Write a program to interpolate the function

$$f(x) = \frac{1}{25x^2 + 1}, \quad x \in [-1, 1],$$

with a polynomial of degree $\leq n$ using the Newton's interpolation formula

- at the equidistant points $x_j = -1 + 2j/n, \; j = 0, 1, \cdots, n.$
- at the zeros $t_k^{(n+1)}, \;\; k = 1, 2, \cdots, n+1$ of the (n+1)-st Chevbyshev polynomial T_{n+1}.

Plot the respective graphs for $n = 5, 10$. Also submit your codes in both paper and email. You could use Language such as C/C++, Fortran, Matlab or Maple.